

LINUX LAB-1

Que-2

Que 2: Demonstrate pipes & redirection (>, >>, <, 2>, 2>>, 2>&1, |) with cat, grep, sort, uniq, wc, tee to generate filtered reports.

Ans 2: Linux provides pipes and redirection operators for transferring and storing command output. The pipe (|) sends the output of one command as the input of another command. The > operator redirects output to a file and overwrites its previous content, while >> appends output to an existing file. The < operator provides input from a file. 2> and 2>> redirect standard error to a file, while 2>&1 combines standard error with standard output. The tee command saves output into a file while also passing it to the next command. Commands such as cat, grep, sort, uniq, and wc can be combined using these operators to generate filtered reports.

Result

The pipes and redirection operations were successfully demonstrated using cat, grep, sort, uniq, wc, and tee.

Filtered data, sorted unique records, line counts, and error reports were successfully generated.

```
root@mshuaibsi: /home/LabF  ×  +  ▾  
root@mshuaibsi:/home/LabFiles# cat > sample_data.txt << 'EOF'  
apple 10  
banana 20  
apple 10  
cherry 30  
banana 20  
EOF  
root@mshuaibsi:/home/LabFiles# cat sample_data.txt | grep apple > apple_report.txt  
root@mshuaibsi:/home/LabFiles# cat sample_data.txt | sort | uniq >> apple_report.txt  
root@mshuaibsi:/home/LabFiles# wc -l sample_data.txt >> apple_report.txt  
root@mshuaibsi:/home/LabFiles# cat sample_data.txt | tee full_copy.txt | wc -l  
5  
root@mshuaibsi:/home/LabFiles# ls -l /nonexistent 2> error_log.txt  
root@mshuaibsi:/home/LabFiles# cat sample_data.txt 1> out.txt 2> err.txt  
root@mshuaibsi:/home/LabFiles# cat sample_data.txt 2>&1 | tee combined.txt > /dev/null  
root@mshuaibsi:/home/LabFiles# ls -l apple_report.txt full_copy.txt error_log.txt out.txt err.txt comb  
ined.txtls -l apple_report.txt full_copy.txt error_log.txt out.txt err.txt combined.txt  
ls: cannot access 'combined.txtls': No such file or directory  
-rw-r--r-- 1 root root 65 Sep 19 16:09 apple_report.txt  
-rw-r--r-- 1 root root 65 Sep 19 16:09 apple_report.txt  
-rw-r--r-- 1 root root 48 Sep 19 16:10 combined.txt  
-rw-r--r-- 1 root root 0 Sep 19 16:10 err.txt  
-rw-r--r-- 1 root root 0 Sep 19 16:10 err.txt  
-rw-r--r-- 1 root root 60 Sep 19 16:10 error_log.txt  
-rw-r--r-- 1 root root 60 Sep 19 16:10 error_log.txt  
-rw-r--r-- 1 root root 48 Sep 19 16:09 full_copy.txt  
-rw-r--r-- 1 root root 48 Sep 19 16:09 full_copy.txt  
-rw-r--r-- 1 root root 48 Sep 19 16:10 out.txt  
-rw-r--r-- 1 root root 48 Sep 19 16:10 out.txt
```